

Mini Test 6

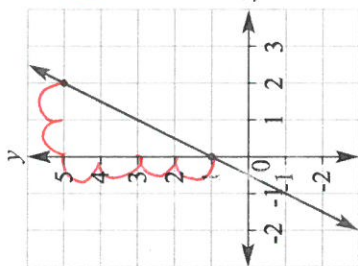
Name : Solution

14

Question 1

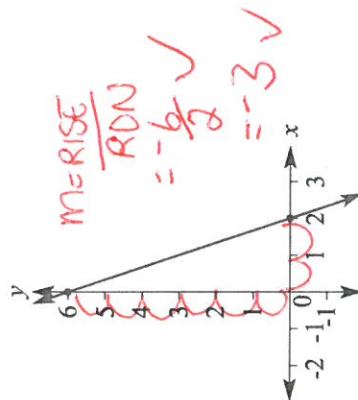
Find the gradient of the following lines.

a



$$m = \frac{\text{RISE}}{\text{RUN}} = \frac{4}{2} = 2$$

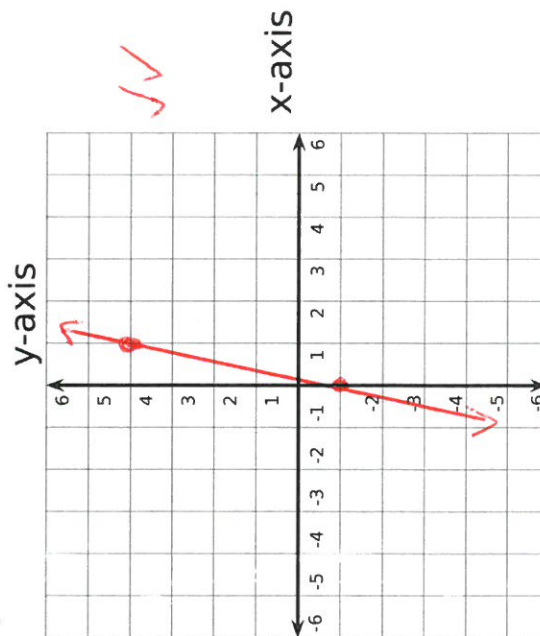
b



$$m = \frac{\text{RISE}}{\text{RUN}} = \frac{6}{3} = 2$$

Question 2

Plot a line with a gradient of 5 that goes through the point (1, 4)



Question 3

Find the gradient of the lines joining the following pairs of points.

a A(2, 3) and B(3, 5)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 3}{3 - 2} = \frac{2}{1} = 2$$

b C(1, 2) and D(3, -4)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 2}{3 - 1} = \frac{-6}{2} = -3$$

Question 4

a) Determine the equation of the straight line with the gradient -2 and y-intercept = -2

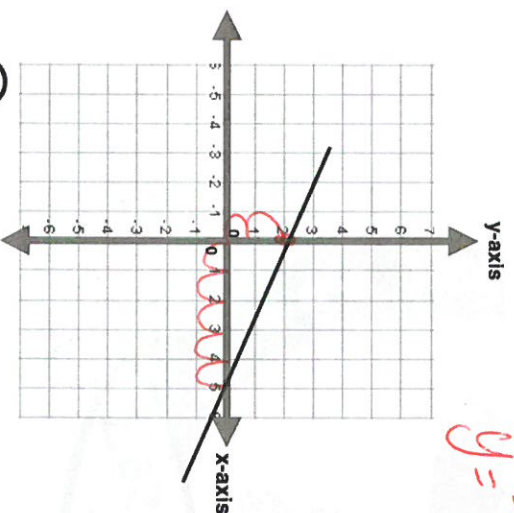
$$y = -2x - 1$$

b) Determine the equation of the straight line with the gradient 5 passing through (0, 6).

$$y = 5x + 6$$

Question 5

What is the equation of this line?



$$y = -\frac{2}{5}x + 2$$